

Galileo automatic measurements campaign

GRAIN

Python automation for GNSS monitoring and coordinated measurement matching

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Reliable measurements start on the ground

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OPERATIONAL CONTEXT

Monitor GNSS signals with a reliable receiving chain.

The campaign combines ground equipment, satellite planning inputs and partner recordings.

CNES

French National Centre for Space Studies



CNES entrance



Toulouse

Ground infrastructure + coordinated data

The GNSS receiving chain

A ground-based path from signal collection to measurement analysis

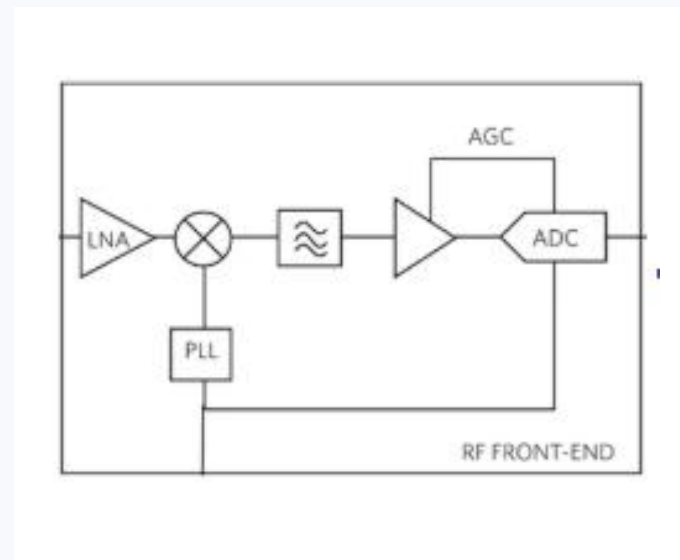
01



Parabolic antenna

Collect the satellite signal

02



RF front-end

Condition the received signal

03



Baseband digitizer

Digitise the received signal

04



Spectrum analyser

Inspect the measured spectrum

The standard GPS campaign

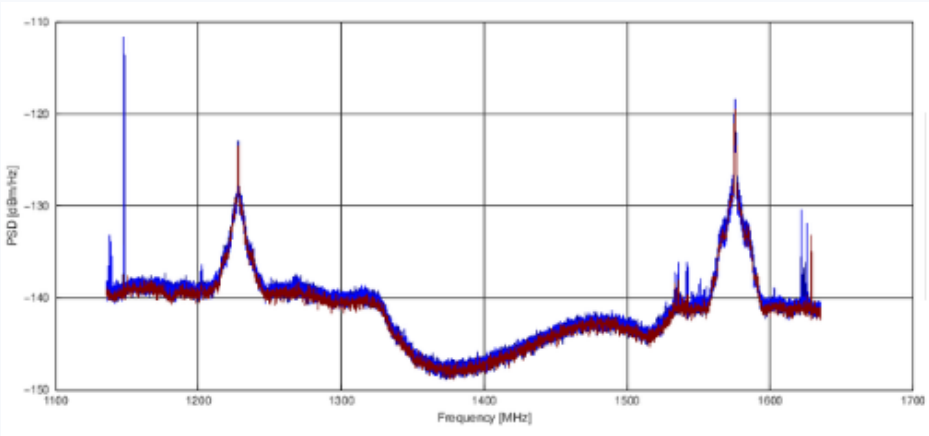
Orbital information becomes antenna commands and measurement data

01 ORBITAL INPUT

Name of satellite										Int. Designation (13=year, A=first item off the launcher)									
				Epoch		Mean motion		Ephemeris type											
						1st deriv		2nd deriv		Drag coeff		Element no							
1	39188U	13031A	14318.21238429	-0.00000028	00000-0	00000+0	0	1302	Check sums										
2	39188	0.0402	340.8502	0003409	258.5822	120.5402	5.00116345	25340											
Satellite number		Right ascension of ascending node		Argument of perigee		Mean anomaly		Mean motion		Revolution no									
Orbit inclination																			



Command file



Two-line elements (TLE)

Orbital parameters used by the campaign planning software

Antenna control

The receiving chain and controller execute the plan.

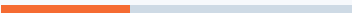
GPS signal band

Original measurement plot

PLAN

COMMAND

ACQUIRE



Align measurements across satellites

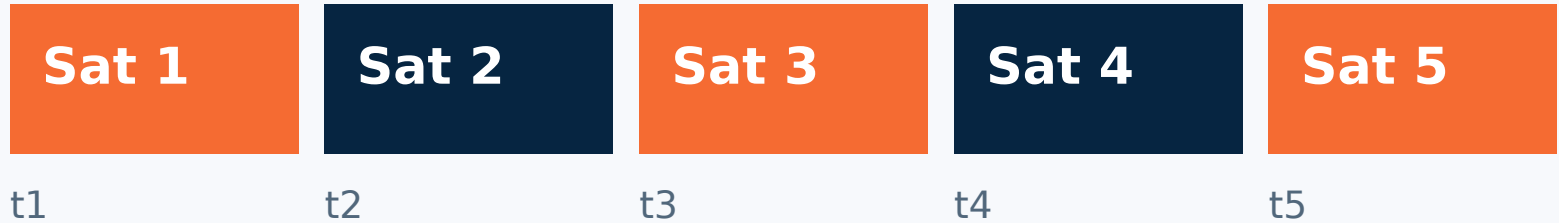
The campaign needs a common timeframe for comparable measurements

GPS REFERENCE CAMPAIGN

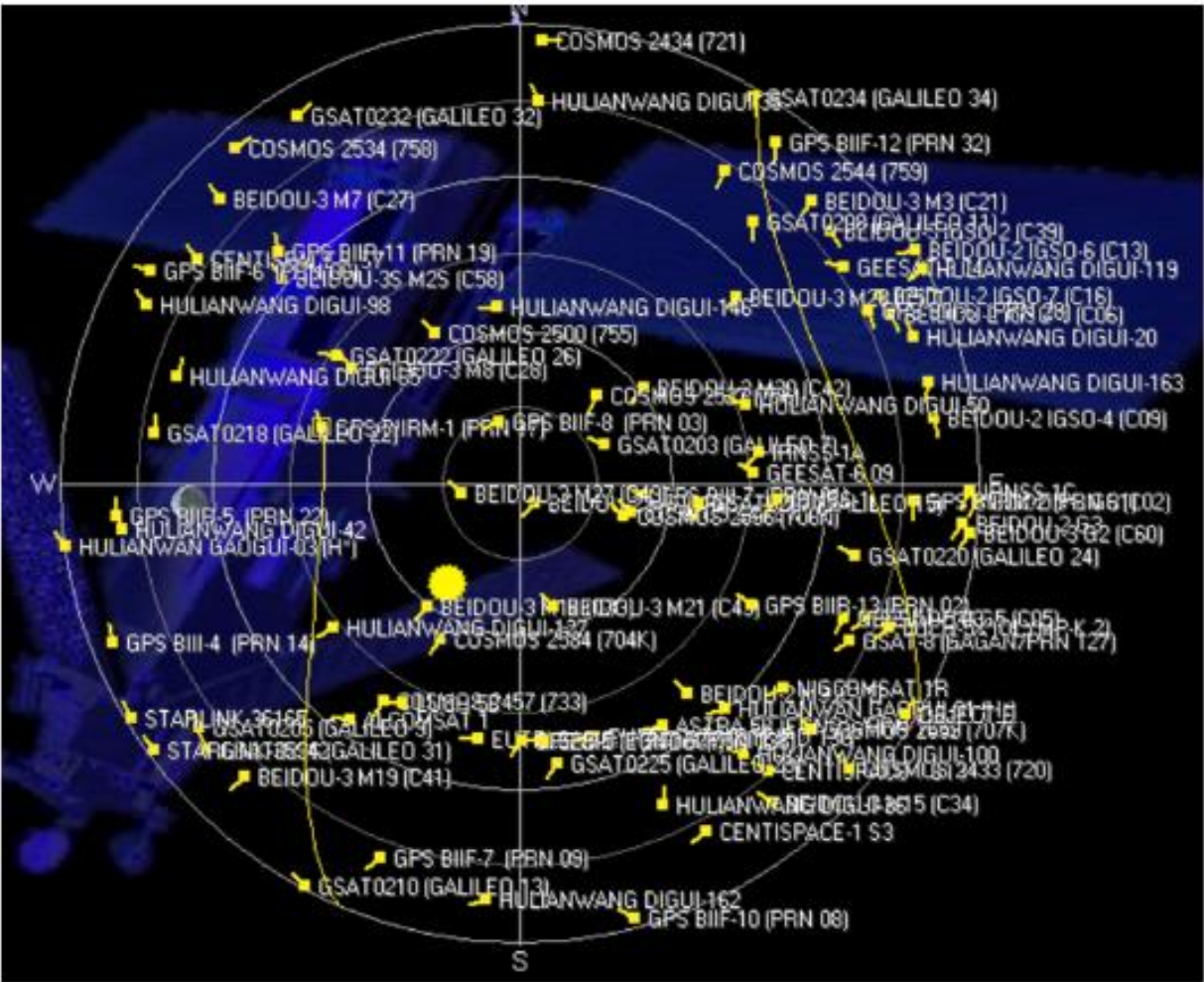


BeginEnd

GALILEO CAMPAIGN



Satellite-specific windows must be coordinated.



Orbitron radar view

From campaign inputs to executable commands

Five stages connect partner measurements with CNES acquisition planning



PYTHON AUTOMATION / Clear inputs, explicit selection rules, traceable outputs

Match the same satellite at the closest time

Satellite identity and temporal proximity make the datasets comparable

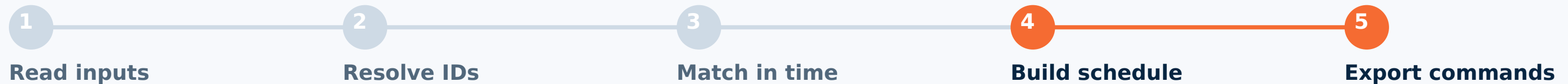


- | | | |
|-------|----------------------------|--|
| 1 | Read & identify | Parse the files and associate satellite names with NORAD IDs. |
| <hr/> | | |
| 2 | Compare | For each partner record, calculate the time gap to CNES candidates. |
| <hr/> | | |
| 3 | Filter & rank | Keep candidates within the configured window; rank by absolute time gap. |
| <hr/> | | |
| 4 | Reserve | Retain nearby candidates and remove them from the CNES candidate pool. |

The candidate pool is then passed to the scheduling stage.

Turn candidate pairs into a usable campaign

Selection rules account for acquisition timing and instrument constraints

**01**

Select recordings

Rank candidate pairs by time gap. Select up to the configured quota for each satellite.

02

Respect time spacing

Enforce a minimum gap between all selected acquisitions, then sort the campaign by time.

03

Generate commands

Export the adapted .col files. Split the schedule into batches of at most 100 recordings.

TRACEABILITY

Intermediate and final CSV tables retain the processing outputs.

Make the campaign easy to check and repeat

Automation supports efficiency, coordination and data reliability



CAMPAIGN CHECKS

Verify constellation coverage

Compare collected NORAD IDs with the reference constellation list.
Identify missing or unexpected satellites.

REVIEWABLE OUTPUT

Summarise the acquisition plan

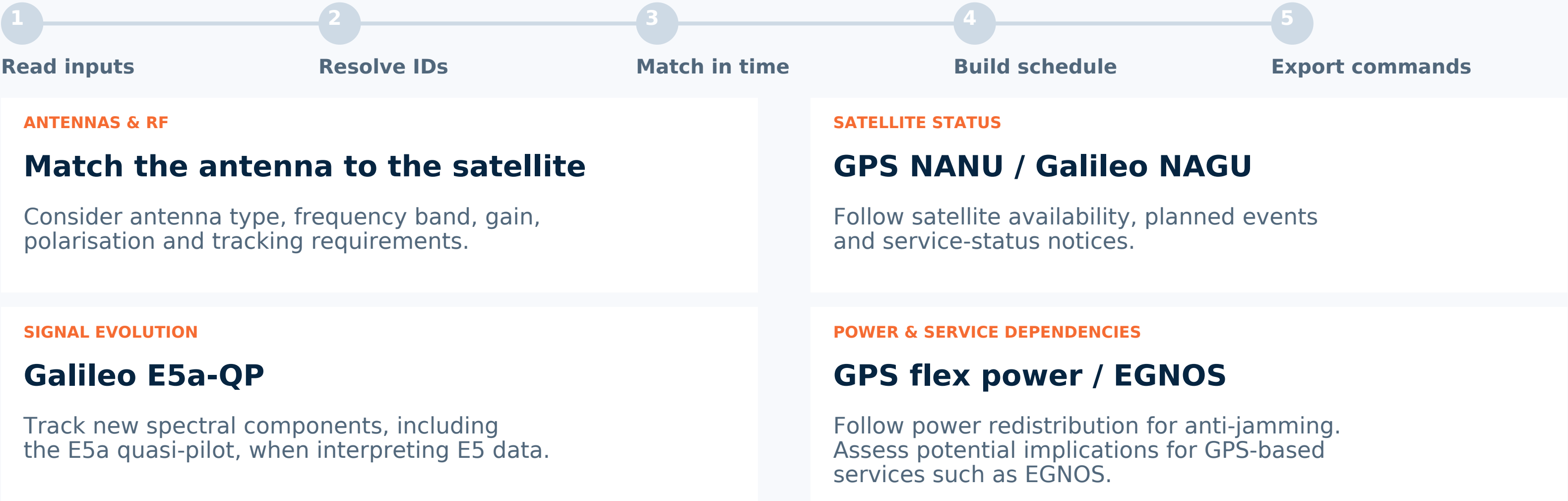
Count recordings per satellite.
Compare declared and parsed totals.
Export a structured Excel summary.

ADDITIONAL AUTOMATION EXPERIENCE

I also developed an automated tool to verify compliance with specifications.

Interpret measurements in their system context

Beyond GRAIN: staying current with hardware, satellite status and signal evolution



References: NAVCEN (NANU) · GSC (NAGU / E5a-QP) · DLR (flex power) · EUSPA (EGNOS)

Campaign coverage and documented outcomes

Results presented to EUSPA and documented in a report



30 / 30

Targeted satellites

Complete constellation coverage

≥ 2

Measurements per satellite

Minimum reported in the campaign

≤ 25 s

Temporal alignment

Absolute CNES-partner time gap

PRESENTATION TO EUSPA

Campaign results communicated

REPORT

Results and methodology documented